

04/01/2026.

→ $f(A) = f(A^T)$

→ If A is full rank $m \times n$ matrix then A^T will also be a full rank matrix if $m \neq n$

→ $A^T A$ will be a RSM of matrix n

$\therefore \text{rank} = n \Rightarrow |A^T A| \neq 0 \Rightarrow$ No eigenvalue is zero

→ $A A^T$ is a RSM of matrix rank $= n$

$(\therefore f(AA^T) = f(A^T A) = f(A) = f(A^T))$

$\therefore \text{rank} = m \Rightarrow |A A^T| = 0 \Rightarrow (m-n)$ eigenvalues are zero

→ $\text{card}(\text{eigenval}(A^T A)) = n$

$\text{card}(\text{eigenval}(A A^T)) = m$, and

Note: the eigenval $(A^T A) = \text{eigenval}(A A^T)$ {why? Later}

Moving to SVD, we can write

$A = U \Sigma V^T$ prove. $U_{m \times m} \Sigma_{m \times n} V_{n \times n}^T$

where

$U = \begin{bmatrix} u_1 & u_2 & \dots & u_f & x_1 & x_2 & \dots & x_{n-f} \end{bmatrix}_{m \times m}$

Dubbi: Σ^+ is also known as Moore Penrose pseudoinverse

$\Sigma = \begin{bmatrix} \sigma_1 & & & & & \\ & \sigma_2 & & & & \\ & & \ddots & & & \\ & & & \sigma_f & & \\ & & & & & 0 \end{bmatrix}_{m \times n}$

$V^T = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_f^T \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{n \times n}$

when we multiply:

$$A = U \Sigma V^T = \lambda_1 u_1 v_1^T + \lambda_2 u_2 v_2^T + \dots + \lambda_3 u_3 v_3^T + 0 + \dots$$

From earlier derivation: $A = U \Sigma V^T = \sum_{i=1}^r \lambda_i u_i v_i^T$ + hence proved.

Given $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, what does A do to x to result in $\bar{b}$?

$$A = \sum_{j=1}^r \lambda_j v_j u_j^T$$

$u_j \in \mathbb{R}^m$
 $v_j \in \mathbb{R}^n$

$$A \bar{x} = \sum_{j=1}^r \lambda_j v_j u_j^T \bar{x} = \sum_{j=1}^r \lambda_j v_j \left(\frac{u_j^T \bar{x}}{u_j^T u_j} \right)$$

Observation:

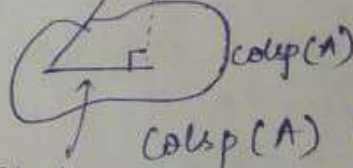
- Project x on the orthonormal basis of $\mathbb{R}^n$.
- Shifts from $\mathbb{R}^n$ domain to $\mathbb{R}^m$.
- Scale that constant vector with λ_j in $\mathbb{R}^m$.

Solve $A \bar{x} = \bar{b}$ where $\bar{b} \notin \text{colsp}(A)$.

No soln exists $\because \bar{b} \notin \text{colsp}(A) \Rightarrow A \bar{x} \neq \bar{b}$

$$\bar{b} - A \bar{x} \neq 0$$

Can we find an approximate soln.



Can we find a soln as close to the $\text{colsp}(A)$ i.e. Reduce the Variance / error?

$A x_{\text{opt}}$

$\|b - Ax\|^2$ as less as possible.

We need a x_{opt} such that $\|b - Ax_{\text{opt}}\|^2 \leq \|b - Ax\|^2$

$A x_{\text{opt}} = Pb$ where P is a matrix which projects b onto the $\text{colsp}(A)$.

$$\text{WKT } (b - Ax_{\text{opt}}) \perp Ay \quad \forall y \in \mathbb{R}^n$$

$$\text{i.e. } (b - Ax_0) \cdot Ay = 0 \quad \forall y \in \mathbb{R}^n$$

$$(Ay)^T (b - Ax_0) = 0$$

$$= y^T (A^T b - A^T A x_{opt}) = 0.$$

$$= y \cdot (A^T b - A^T A x_0) = 0 \quad \forall y \in \mathbb{R}^n$$

$$\Rightarrow A^T b - A^T A x_{opt} = \vec{0}. \quad A_{m \times n}$$

$$A^T A x_0 = A^T b \quad \forall y \in \mathbb{R}^n \quad A^T A_{n \times n}$$

$$x_0 = (A^T A)^{-1} A^T b.$$

$$\text{Pseudo Inverse} = A^{\dagger} = (A^T A)^{-1} A^T.$$

This inverse is possible only if A is full rank
because if A is full rank $\Rightarrow \text{rank}(A) = n$

$$\Rightarrow \text{rank}(A^T A_{n \times n}) = n \Rightarrow (A^T A) \text{ is invertible}$$

$$\text{Now, } A^{\dagger} = (A^T A)^{-1} A^T$$

Remember we told $A x_0 = P b$.
What is P ?

is also called
Gram Matrix if
 $A^T A$ is invertible

Substitute,

$$A (A^T A)^{-1} A^T b = P b.$$

Gilbert Strang

$$\Rightarrow P = \text{Projection Matrix} = A (A^T A)^{-1} A^T = A A^{\dagger}$$

↳ Idempotent: $P^2 = P$
↳ Symmetric: $P^T = P$

Now, let's find $A^{\dagger} A A^{\dagger}$

$$= [(A^T A)^{-1} A^T] A [(A^T A)^{-1} A^T]$$

$$= (A^T A)^{-1} (A^T A) [(A^T A)^{-1} A^T] = A^{\dagger}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

$$A^T A_{3 \times 3}$$

$$A^T_{2 \times 2}$$

i.e. for $A^{m \times n}$, $f(A) = m$.
 $\therefore f(AA^T) = 2$

$\therefore$ we can write.

$$(AA^T)(AA^T)^+ = I$$

$$A(\underbrace{A^T(AA^T)^+}_{\text{pseudo inverse of } A^T}) = I$$

Observation:

- For 3×2 matrix: $A^+ = A^T(AA^T)^+$

- For 2×3 matrix: $A^+ = (A^T A)^+ A^T$

Now what if $A^{n \times n}$ with rank = n .

$$AA^+ = A(A^T A)^+ A^T$$

$$= A(A^+)(A^T)^+ A^T \quad \because \text{all } A, A^T, A^+ \text{ are invertible}$$

$$= I I = I$$

$\therefore A$ is full rank.

$\therefore$ For invertible square matrix: $A^+ = A^{-1}$.

$$A \cdot A^+ \cdot A = A$$

- If A is a projection matrix.

$$A^2 = A$$

$$A(A) = A$$

$$A^+ = A$$

$$(A^T A)^+ A^T$$

$$(A^2)^+ A$$

$$A^+ A = A$$

Welcome back to 6480.

Input : $\{u, v, w\}$

Output $\bar{e} = \bar{u}$

$$\bar{f} = \bar{v} - \left(\frac{\bar{v} \cdot \bar{e}}{\bar{e} \cdot \bar{e}} \right) \bar{e}$$

$$\bar{g} = \bar{w} - \left(\frac{\bar{w} \cdot \bar{e}}{\bar{e} \cdot \bar{e}} \right) \bar{e} - \left(\frac{\bar{w} \cdot \bar{f}}{\bar{f} \cdot \bar{f}} \right) \bar{f}$$

To express a matrix.

$$[u \ v \ w] = [e \ f \ g] \begin{bmatrix} 1 & \frac{v \cdot e}{e \cdot e} & \frac{w \cdot e}{e \cdot e} \\ 0 & 1 & \frac{w \cdot f}{f \cdot f} \\ 0 & 0 & 1 \end{bmatrix}$$

VI make sure u, v, w are orthonormal vectors.

$A = QR$ only if A has all linearly independent columns.

$$A_{m \times n} = \underbrace{Q_{m \times n}}_{\text{orthogonal matrix}} \underbrace{R_{n \times n}}_{\text{upper triangle matrix}}$$

if $m \neq n$. $A^T = (QR)^T = R^T Q^T$. But $\because A_{m \times n}$ Q^T not exists $\therefore R^T Q^T$.

what is $AA^T = Q R R^T Q^T = QQ^T$

Perform QR decomposition of $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$.
 $\begin{matrix} u & v \end{matrix}$

$$\vec{e} = \vec{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \vec{f} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \left(\frac{0+0}{+1} \right) \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\vec{e} = \vec{u} = \frac{1}{\sqrt{2}} [1 \ 0 \ 1]^T$$

$$\vec{f} = \frac{1}{\sqrt{2}} [0 \ 1 \ 1]^T - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} [1 \ 0 \ 1]$$

$$\vec{v} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{\sqrt{2}-1}{2} \end{bmatrix}$$

$$\vec{e} \cdot \vec{f} = \frac{-1}{2\sqrt{2}} + 0 + \frac{(\sqrt{2}-1)}{2\sqrt{2}}$$

For a matrix det all eigenvalues of a matrix. be = $\vec{e}$.

→ If $E > 0$ positive definite
 $E < 0$ negative definite
 $E \geq 0$ true semi definite.
 $E \leq 0$ -ve semi definite
 $E = 0$ Indefinite.

Orthonormal basis

Let $\forall \bar{u} \in V$ of $\dim \mathbb{R}^k$

$$\bar{u} = \frac{\bar{u} \cdot \phi_1}{\phi_1 \cdot \phi_1} \phi_1 + \frac{\bar{u} \cdot \phi_2}{\phi_2 \cdot \phi_2} \phi_2 + \dots + \frac{\bar{u} \cdot \phi_k}{\phi_k \cdot \phi_k} \phi_k$$

where $S = \{\phi_1, \phi_2, \dots, \phi_k\}$ are orthonormal vectors.
 $\phi_i \cdot \phi_i = 1$ (if orthonormal)

Prerequisites:

1 fundamental subspaces of a matrix.

Orthonormal basis

* Zero vector is orthogonal to every other vector for a given subspace.

Orthogonal Complement Subspace.

Let $V = \mathbb{R}^n$ if W is subspace of $\mathbb{R}^n$ of $\dim k$, then

$$W^\perp = \{ \vec{v} \in \mathbb{R}^n \mid \vec{v} \cdot \vec{w} = 0 \ \forall \vec{w} \in W \}$$

Properties

1) $W \cap W^\perp = \{ \vec{0} \}$ $\neq$ Proof. Let $\vec{v} \in W \cap W^\perp$
 then $\vec{v} \in W$ as well as $\vec{v} \in W^\perp$
 which means $\vec{v} \cdot \vec{v} = 0$
 Only magnitude of $\vec{0}$ is 0.

2) Let $V = \mathbb{R}^n$ and $W =$ subspace of V

Let basis of $V = \{u_1, u_2, \dots, u_k, u_{k+1}, \dots, u_n\}$

Let $\{u_1, u_2, \dots, u_k\}$ be ONB for W ,

Then $\{u_{k+1}, u_{k+2}, \dots, u_n\}$ is a ONB for $W^\perp$.

Proof:

Given: $V = \text{span}(\{u_1, u_2, \dots, u_k, u_{k+1}, \dots, u_n\})$ ONB — ①
 $W = \text{span}(\{u_1, u_2, \dots, u_k\})$ — ②

To prove: $W^\perp = \text{span}(\{u_{k+1}, u_{k+2}, \dots, u_n\})$

Let $S = \text{span}(\{u_{k+1}, \dots, u_n\})$

then we can prove $W^\perp = S$ if $S \subseteq W^\perp$ and $W^\perp \subseteq S$.

Case 1: To prove $S \subseteq W^\perp$

Any vector $\bar{x} \in S = d_{k+1} \bar{u}_{k+1} + \dots + d_n \bar{u}_n$ for some scalars

Any vector $\bar{w} \in W = c_1 \bar{u}_1 + \dots + c_k \bar{u}_k$ for some scalars

From ① $\bar{x} \cdot \bar{w} = 0$, because every vector in $\{u_1, u_2, \dots, u_n\}$ is

obviously orthonormal to every other vector in that set.

$$\therefore \bar{x} \in W^\perp \Rightarrow S \subseteq W^\perp$$

Case 2: To prove $W^\perp \subseteq S$

For any $\bar{x} \in W^\perp = (\bar{x} \cdot \bar{u}_1) \bar{u}_1 + (\bar{x} \cdot \bar{u}_2) \bar{u}_2 + \dots + (\bar{x} \cdot \bar{u}_k) \bar{u}_k +$
 $(\bar{x} \cdot \bar{u}_{k+1}) \bar{u}_{k+1} + \dots + (\bar{x} \cdot \bar{u}_n) \bar{u}_n$

Now if $\bar{x} \in W^\perp$ then from (2)

$$\bar{x} = (\bar{x} \cdot \bar{u}_{k+1}) \bar{u}_{k+1} + \dots + (\bar{x} \cdot \bar{u}_n) \bar{u}_n$$

$$\bar{x} \in \text{span}(\{\bar{u}_{k+1}, \dots, \bar{u}_n\}) \in S$$

$$\therefore \bar{x} \in S$$

From case (1) and (2) $W^\perp = S$.

(A nice beginning to Gram's expansion)

$$3) \rightarrow \dim(W) + \dim(W^\perp) = \dim(V)$$

4) Let $W = \text{span}(\text{ONB}(\bar{u}_1, \bar{u}_2, \dots, \bar{u}_k))$ and $W \subseteq \mathbb{R}^n$

For any vector $\bar{v} \in \mathbb{R}^n$ we define, the orthogonal projection of $\bar{v}$ on W as

$$\text{proj}_W \bar{v} = (\bar{v} \cdot \bar{u}_1) \bar{u}_1 + (\bar{v} \cdot \bar{u}_2) \bar{u}_2 + \dots + (\bar{v} \cdot \bar{u}_k) \bar{u}_k$$

5) For any vector $\bar{v} \in \mathbb{R}^n$ we can express $\bar{v}$ uniquely as

$$\bar{v} = \bar{v}_W + \bar{v}_{W^\perp} \text{ where } \bar{v}_W \in W, \bar{v}_{W^\perp} \in W^\perp$$

and since $\bar{v} = \bar{v}_W + \bar{v}_{W^\perp}$ then

$$\|\bar{v}\|^2 = (\bar{v}_W + \bar{v}_{W^\perp}) \cdot (\bar{v}_W + \bar{v}_{W^\perp})$$

$$\|\bar{v}\|^2 = \|\bar{v}_W\|^2 + \|\bar{v}_{W^\perp}\|^2 \quad (\text{Pythagoras Theorem})$$

For proving property (2) we took into account that

$$\dim(W) = k + \dim(W^\perp) = n - k = \dim(V) = n$$

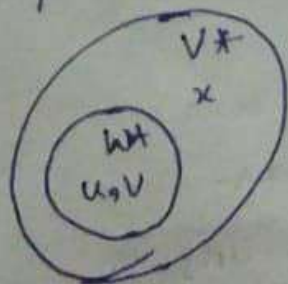
But now to find $\dim(W^\perp)$?

Another way to solve this is the Gram's expansion Theorem.

3) Gram's Expansion Theorem -

Given: $\{u, v\}$ are linearly independent $\&$ $u, v \in W^*$

To prove: $\{u, v, x\}$ are linearly independent where $x \in V^* - W^*$



Given: $\{u, v\}$ are L.I.

$$\therefore a\bar{u} + b\bar{v} = \bar{0} \Leftrightarrow a = b = 0$$

If $\bar{u}, \bar{v}, \bar{x}$ are L.I. then

$$\alpha\bar{u} + \beta\bar{v} + \lambda\bar{x} = \bar{0}$$

$$\text{If } \lambda = 0 \Rightarrow \alpha\bar{u} + \beta\bar{v} = \bar{0} \quad \checkmark$$

$$\text{If } \lambda \neq 0 \Rightarrow \bar{x} = -\frac{\alpha}{\lambda}\bar{u} - \frac{\beta}{\lambda}\bar{v}$$

- This means $\bar{x}$ can be expressed in terms of u, v .
- which means $\bar{x}$ must $\in W^*$
- but given $\bar{z} \in V^* - W^*$
- which means λ has to be zero.
- which means $\alpha \bar{u} + \beta \bar{v} + \lambda \bar{z} = 0 \Leftrightarrow \alpha = \beta = \lambda = 0$
- which means $\{\bar{u}, \bar{v}, \bar{z}\}$ are LI.

$\subseteq$ means subspace.

NOTE-

- No of components of a vector need not be equal to dimension of Vector subspace.
- any set of $(n+1)$ vectors in $\mathbb{R}^n$ always form a linearly dependent set. This is not true for all Vector spaces.
- Eg: For a Vector space consisting of polynomials of degree $\leq n$, the number of basis vectors itself is $(n+1)$.

4) Orthogonal Basis Expansion Theorem.

Previous proof was to prove Linear Independence. Now,

Question: Let $W \subseteq \mathbb{R}^K$ | $\dim(W) = d$

Let $W^\perp =$ orthogonal complement of W .

If $W = \text{span}(\text{ONB}(\phi_1, \phi_2 \dots \phi_d))$ and $W^\perp = \text{span}(\psi_1, \psi_2 \dots \psi_{n-d})$
can we extend the ONB of W to form a basis for $\mathbb{R}^K$.

Answer: Since, $\dim(\mathbb{R}) = K$ we need $(n-d)$ vectors more.

Let these vectors be $\{\psi_1, \psi_2 \dots \psi_{n-d}\}$ orthogonal vectors.

$\therefore$ To prove: $\forall \bar{x} \in V, \bar{x} = \text{span}(\text{ONB}(\phi_1, \phi_2 \dots \phi_d, \psi_1, \psi_2 \dots \psi_{n-d}))$

- From previous theorem we can tell $\{\phi_1, \dots \phi_d, \psi_1, \dots \psi_{n-d}\}$ are linearly independent. ——— ①

- And, since there are $d + (n-d) = n$, LI vectors they can span $\mathbb{R}^K$.

Question: Now if $W = \text{span}(\text{ONB}(\phi_1, \phi_2 \dots \phi_d))$ can we tell

$W^\perp = \text{span}(\text{ONB}(\psi_1, \psi_2 \dots \psi_{n-d}))$.

Proof: Let $\bar{x} \in \mathbb{R}^n$. $\therefore \bar{x} = \text{span}(\text{ONB}(\phi_1, \phi_2 \dots \phi_d, \psi_1, \dots \psi_{n-d}))$

which can be written as $\bar{x} = \sum_1^d (\phi_i \cdot \bar{x}) \bar{\phi}_i + \sum_1^{n-d} (\psi_i \cdot \bar{x}) \bar{\psi}_i$

Now if $\bar{x} \in W^\perp$ then $\bar{x} = 0 + \sum_1^{n-d} (\psi_i \cdot \bar{x}) \bar{\psi}_i$
 $\therefore \phi_i \cdot \bar{x} = 0$
 if $\bar{x} \in W^\perp$

which can be written as $\bar{x} = \text{span}(\text{ONB}(\psi_1, \dots, \psi_{n-d}))$ for $x \in W$.
 So we have effectively proved that

From ① and ②
 If $W = \text{span}(\text{ONB}(\phi_1, \phi_2, \dots, \phi_d))$, and

$W^\perp = \text{span}(\text{ONB}(\psi_1, \psi_2, \dots, \psi_{n-d}))$, then the

basis of $W^\perp$ can be expanded using W to span $\mathbb{R}^K$.
 which means $\forall x \in \mathbb{R}^K$, $\bar{x} = \underbrace{\sum_{j=1}^d (x \cdot \phi_j) \phi_j}_{\text{Some vector in } W} + \underbrace{\sum_{j=d+1}^{n-d} (x \cdot \psi_j) \psi_j}_{\text{Some vector in } W^\perp}$

Wow! Now we have also proved that $\forall x \in \mathbb{R}^K$,
 x can be projected (or decomposed) as
 $\bar{x} = x_W + x_{W^\perp}$ where $x_W \in W$ and $x_{W^\perp} \in W^\perp$

Question: Is this decomposition unique or Non unique?
 i.e. for a given $\bar{x} \in \mathbb{R}^K$ is $\bar{x}$ the only vector which
 which can be written as $\bar{x} = x_W + x_{W^\perp}$ OR
 does some other vector ~~exists~~ exists which can
 be written as $\bar{x} = x_W^* + x_{W^\perp}^*$

Proof: Let $\bar{x} \in \mathbb{R}^K$ have 2 decompositions.

$$\bar{x} = \bar{x}_W + \bar{x}_{W^\perp} \text{ , and}$$

$$\bar{x} = x_W^* + x_{W^\perp}^*$$

$$\text{Subtract: } \bar{x} - \bar{x} = 0 = (\bar{x}_W - x_W^*) + (x_{W^\perp} - x_{W^\perp}^*)$$

$$\Rightarrow x_W = x_W^* \text{ and } x_{W^\perp} = x_{W^\perp}^*$$

Another way of observing this:

$$0 = \underbrace{(\bar{x}_W - x_W^*)}_{\in W} + \underbrace{(x_{W^\perp} - x_{W^\perp}^*)}_{\in W^\perp} \quad \text{--- ①}$$

$$(\bar{x}_W - x_W^*) = -(x_{W^\perp} - x_{W^\perp}^*) \quad \text{--- ②}$$

From ① and ② we can conclude that

$$(\bar{x}_W - x_W^*), (x_{W^\perp} - x_{W^\perp}^*) \in W \text{ as well as } W^\perp = W \cap W^\perp$$

But we know (from property 2.1) that $W \cap W^\perp = \{0\}$

$$\therefore \bar{x}_W - x_W^* = 0 \Rightarrow \bar{x}_W = x_W^*$$

$$x_{W^\perp} - x_{W^\perp}^* = 0 \Rightarrow x_{W^\perp} = x_{W^\perp}^*$$

Question: What is the orthogonal complementary subspace of N_A and C_{A^T} ?

Answer: For $x_{n \times 1} \in N_A \Rightarrow A_{m \times n} x_{n \times 1} = \vec{0}_{m \times 1}$
For any vector $y \in \mathbb{R}^m$ we can write

$$\vec{y}_{m \times 1} \cdot \vec{0}_{m \times 1} = 0.$$

$$\vec{y}_{m \times 1} \cdot (A_{m \times n} x_{n \times 1})_{m \times 1} = 0.$$

$$\text{WKT } a \cdot b = a^T b = b^T a.$$

$$\therefore \text{ we can write as } (Ax)^T y = 0.$$

$$x^T A^T y = 0.$$

$$(x^T)_{1 \times n} (A^T y)_{n \times 1} = 0.$$

Observation:

- we have a vector $A^T y$ which on dot product with x gives 0
- If $a \cdot b = 0$, it means $\{a, b\}$ are ON vectors
- For $x \in N_A$ we have $A^T y = 0$, where $y \in \mathbb{R}^m$
- $A^T y$ for $y \in \mathbb{R}^m$ is actually the colspace (A^T) .
- ⇒ $A^T y$ (i.e. C_{A^T}) and x (i.e. N_A) are orthogonal complement vectors.

Similarly we can prove C_A and N_{A^T} are orthogonal complement subspaces.

Now, WKT: $\dim(W) + \dim(W^\perp) = \dim(\mathbb{R}^n)$.

$$\therefore \text{ we can write: } \underbrace{\dim(\text{Colsp}(A^T)) + \dim(\text{Nulsp}(A))}_{\text{Rank Nullity theorem through Orthogonal complement subspaces}} = n.$$

$$A = n$$

2) Rank of outer product of 2 vectors always gives matrix of rank 1.

Proof: If $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ then $\vec{v}\vec{v}^T = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$

∴ $\vec{v}\vec{v}^T = \begin{bmatrix} v_1 \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} & v_2 \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} & v_3 \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \end{bmatrix}$ clearly we can observe that there is only one linearly independent column.

Also we can observe $\vec{v}\vec{v}^T$ is a symmetric matrix.

Question: we have seen a proj. matrix P to project a vector onto the direction of its parent vector. But how to find the projection matrix which projects a vector onto a 'plane' spanned by 2 orthogonal vectors, say, $\vec{u}$ and $\vec{v}$.

Soln: How to project a vector $\vec{x}$ onto the plane spanned by orthogonal vectors $\vec{v}$.

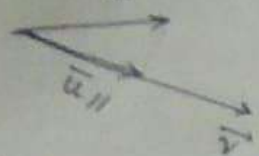
Proof: $\text{Proj}_{(\vec{u}, \vec{v})} \vec{x} = \left(\frac{\vec{u}\vec{u}^T}{\vec{u}^T\vec{u}} + \frac{\vec{v}\vec{v}^T}{\vec{v}^T\vec{v}} \right) \vec{x}$.

NPTEL

∴ Projection matrices are Rank Deficient ($r=1$) they are non invertible.

4) System of Linear Equations

6) Projection



$$\begin{aligned} \bar{u}_{||} &= \left(\frac{\bar{u} \cdot \bar{v}}{\bar{v} \cdot \bar{v}} \right) \bar{v} \\ &= \left(\frac{\mathbf{v}^T \mathbf{u}}{\mathbf{v}^T \mathbf{v}} \right) \bar{v} \end{aligned}$$

Now let

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \text{ and } \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\bar{u}_{||} = \frac{(u_1 v_1 + u_2 v_2)}{v_1^2 + v_2^2} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 u_1 v_1 + v_1 u_2 v_2 \\ v_2 u_1 v_1 + v_2 u_2 v_2 \end{bmatrix} = \begin{bmatrix} u_1 v_1^2 & u_2 v_1 v_2 \\ u_1 v_1 v_2 & u_2 v_2^2 \end{bmatrix}$$

For time being

Let's take $\mathbf{v}$ = unit vector

$$= \begin{bmatrix} v_1^2 & v_1 v_2 \\ v_1 v_2 & v_2^2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \left(\frac{\mathbf{v} \mathbf{v}^T}{\mathbf{v}^T \mathbf{v}} \right) \bar{u}$$

Observation:

→ Outer product of $\mathbf{v}$

= $\mathbf{v} \mathbf{v}^T$

= Projection matrix of $\mathbf{v}$.

$$\text{Projection matrix of } \mathbf{v} = \mathbf{v} \mathbf{v}^T$$

→ Outer product of $\mathbf{v}_{m \times 1}$ and $\mathbf{u}_{1 \times n}$ is $\mathbf{v} \mathbf{u}^T$.

Outer product matrix of $\mathbf{v}$ projects the vector $\mathbf{u}$ in the direction of $\mathbf{v}$.

Properties

1) → If a matrix $\mathbf{P}$ is symmetric ($\mathbf{P}^T = \mathbf{P}$) and Idempotent ($\mathbf{P}^2 = \mathbf{P}$) then it is a projection matrix.

Proof: For a proj matrix $\mathbf{P}_{n \times n}$. If transformed on $\mathbf{b}$, i.e.

For a proj matrix $\mathbf{P}_{n \times n}$ (formed using some vector say $\bar{\mathbf{v}}$)

$\mathbf{P}_{n \times n} \mathbf{b}_{n \times 1}$ will project that vector $\mathbf{b}$ onto the direction of $\bar{\mathbf{v}}$.

i.e. $\mathbf{P}\mathbf{b} = \alpha \bar{\mathbf{v}}$.

→ This implies $\bar{\mathbf{b}} - \mathbf{P}\bar{\mathbf{b}}$ is $\perp$ to $\bar{\mathbf{v}}$.

$\mathbf{P}\bar{\mathbf{b}}$ corresponds to $\text{colsp}(\mathbf{P})$. So we can tell $\bar{\mathbf{b}} - \mathbf{P}\bar{\mathbf{b}}$ is $\perp$ to vectors in $\text{colsp}(\mathbf{P})$.

→ Now any vector in $\text{colsp}(\mathbf{P})$ can be given as $\mathbf{P}\bar{\mathbf{u}}$.

$$\therefore \mathbf{P}\bar{\mathbf{u}} = (\bar{\mathbf{b}} - \mathbf{P}\bar{\mathbf{b}}) = \mathbf{0}$$

$$\mathbf{u}^T \mathbf{P}^T (\bar{\mathbf{b}} - \mathbf{P}\bar{\mathbf{b}}) = 0$$

$$\mathbf{u}^T (\mathbf{P}^T \bar{\mathbf{b}} - \mathbf{P}^T \mathbf{P} \bar{\mathbf{b}}) = 0 \rightarrow \text{Now given } \mathbf{P}^T = \mathbf{P}, \mathbf{P}^2 = \mathbf{P}$$

$$\mathbf{u}^T (\bar{\mathbf{b}} - \mathbf{P}\bar{\mathbf{b}}) = 0$$

$$\text{LHS} = \text{RHS}$$

Supposing

∴ Our assumption is valid.

∴ $\mathbf{P}$ is a projection matrix.

$$A \left(\sum_1^{\lambda} \alpha_i \phi_i \right) + A \left(\sum_1^{n-\lambda} \beta_i u_i \right) = \vec{0}_{m \times 1}$$

$$\underbrace{\sum_1^{\lambda} \alpha_i A_{m \times n} \phi_i}_{\text{vector } n \times 1} + A \left(\sum_1^{n-\lambda} \beta_i u_i \right) = \vec{0}_{m \times 1}$$

$$= 0 \text{ because } A\vec{x} = 0 \text{ for any } \vec{x} \in N_A \text{ — (2)}$$

$$A \left(\underbrace{\sum_1^{n-\lambda} \beta_i u_i}_{\text{vector } n \times 1} \right) = \vec{0}_{m \times 1} \text{ — (3)}$$

If this (3) is true, then from (2) we can write that $\sum_1^{n-\lambda} \beta_i u_i \in N_A$. But this is a contradiction, because our assumption was

$$\text{Let } \text{vec}(N_A) = \text{basis}(N_A) = \{\phi_1, \phi_2, \dots, \phi_\lambda\} \text{ and}$$

$$\text{vec}(C_A) = \{u_1, u_2, \dots, u_{n-\lambda}\}$$

which implies $\text{vec}(C_A) \not\subset \text{vec}(N_A)$.

From proof by contradiction, ~~$\{\phi_1, \phi_2, \dots, \phi_\lambda\}$~~ is a set of LI $\in \text{Colsp}(A^T)$.

(B) Can $\{\phi_1, \dots, \phi_\lambda, u_1, \dots, u_{n-\lambda}\}$ span $\mathbb{R}^n$?

(Shortcut: You can use basis expansion theorem).

Proof: Now we have

$$\text{(Method 2)} \quad N_A = \text{span}(\{\phi_1, \phi_2, \dots, \phi_\lambda\}) \text{ and}$$

$$C_A = \text{span}(\{u_1, u_2, \dots, u_{n-\lambda}\}), \text{ WKT}$$

$$\text{vec}(C_A) \not\subset \text{vec}(N_A) \text{ except } \vec{0}.$$

Using the proof of Basis expansion.

$\therefore \text{vec}(C_A) \not\subset \text{vec}(N_A)$, the set of LI vectors (i.e. basis(N_A)) can be expanded using another set of LI vectors (i.e. basis(C_A)) to span $\mathbb{R}^n$.

$$\star \forall x \in \mathbb{R}^n \Rightarrow \vec{x} = \sum_1^{\lambda_A} \alpha_i \bar{\phi}_i + \sum_1^{n-\lambda_A} \beta_j \bar{u}_j$$

Question: How to determine λ_A ?

WKT: $\text{Rank}(A) = f_A = \text{No. of LI rows} = \text{No. of LI cols in } A$

WKT: $\dim(\text{Colsp}(A)) = n - \lambda_A = \text{No. of LI cols in } A$

$$\therefore \text{Rank}(A) = \dim(\text{Colsp}(A)).$$

$$f_A = n - \lambda_A$$

$$\lambda_A + f_A = n \Rightarrow \text{Rank-Nullity Theorem.}$$

$$\text{Similarly: } \lambda_{A^T} + f_{A^T} = m.$$

Recap.

- ONB, Orthogonal complement subspace.
- Independent basis & ONB expansion theorem.
- properties of orthogonal complement subspaces.

5) Orientation of 4 Fundamental subspaces

Let $A_{m \times n} \in \mathbb{R}^{m \times n}$

Let $c_1, c_2, \dots, c_n = \text{cols}(A)$

Let $d_1, d_2, \dots, d_m = \text{cols}(A^T) = \text{rows}(A)$.

We know $c_i \text{ shape} = m \times 1$, $d_i \text{ shape} = n \times 1$

$\text{Colsp}(A) = \text{span}(c_1, c_2, \dots, c_n) \quad m \times 1$

$\text{Colsp}(A^T) = \text{span}(d_1, d_2, \dots, d_m) \quad n \times 1$

Let $\text{Nullsp}(A) = \text{span}(\phi_1, \phi_2, \dots, \phi_n)$.

Now, wkt, $\text{Colsp}(A) = A_{m \times n} x_{n \times 1} \quad \forall x_{n \times 1} \in \mathbb{R}^n$

$\text{Colsp}(A^T) = A_{n \times m}^T x_{m \times 1} \quad \forall x_{m \times 1} \in \mathbb{R}^m$.

$\text{Nullsp}(A) = x_{n \times 1}$ for which $A_{m \times n} x_{n \times 1} = \vec{0}_{m \times 1}$

$\text{Nullsp}(A^T) = x_{m \times 1}$ for which $A_{n \times m}^T x_{m \times 1} = \vec{0}_{n \times 1}$.

Therefore we can tell:

vectors $(\text{Colsp}(A))$ and vectors $(\text{Nullsp}(A^T)) \in \mathbb{R}^n$

vectors $(\text{Colsp}(A^T))$ and vectors $(\text{Nullsp}(A)) \in \mathbb{R}^m$

Question: For any vector x in $\mathbb{R}^n$ can we tell that

$$\bar{x} = \text{span}\{\text{vec}(\text{Col}(A^T)) \cup \text{vec}(\text{Nullsp}(A))\} \quad \text{--- ①}$$

Proof If statement ① is true then

(A) ~~span~~ $\{\text{vec}(CA^T) \cup \text{vec}(NA)\}$ has to be LI set.

(B) $\text{span}\{\text{vec}(CA^T) \cup \text{vec}(NA)\} = \mathbb{R}^n$.

Let $\text{vec}(NA) = \text{basis}(NA) = \{\phi_1, \phi_2, \dots, \phi_\lambda\}$.

$\therefore$ Remaining vectors needed $= n - \lambda$.

$\therefore$ Let $\text{vec}(CA^T) = \{u_1, u_2, \dots, u_{n-\lambda}\}$.

(A) To prove $\{\phi_1, \phi_2, \dots, \phi_\lambda, u_1, u_2, \dots, u_{n-\lambda}\}$ is a LI set.

To prove $\alpha_1 \phi_1 + \dots + \alpha_\lambda \phi_\lambda + \beta_1 u_1 + \beta_2 u_2 + \dots + \beta_{n-\lambda} u_{n-\lambda} = \vec{0}$

only for $\alpha_1 = \dots = \alpha_\lambda = \beta_1 = \dots = \beta_{n-\lambda} = 0$.

$$\sum_1^\lambda \alpha_i \phi_i + \sum_1^{n-\lambda} \beta_i u_i = \vec{0}_{n \times 1} \quad \text{Multiply by } \star$$